Human Label Variation and Pluralistic Alignment

Natural Language Processing (NDAK18000U)

Course at the University of Copenhagen

20. Oct. 2026

Shanshan Xu

Outline

- Background and Motivation
- What is Human Label Variation (HLV)
- Split Vote in ECtHR as a case study of HLV in Legal NLP

Human Label Variation (HLV)

- Variation/Disagreement in Human Annotation
- Much NLP progress relies on annotated datasets
- Assume **single gold label per instance, e.g.**
    - **Is Berlin the capital of Germany?**
        - **(Human Annotator 1/2/3:** "yes"/ "yes"/ "yes"**)**

Single gold label:
Disagreement in annotation/HLV regarded as noise to be discarded
Dataset quality depends on inter annotator agreement (IAA)

Human Label Variation (HLV)

- **However, in complicated/ subjective/ ambiguous tasks,** a single "ground truth" often does not hold
  - Premis: Mary drinks oat milk
  - Hypothesis: Mary drinks milk
  - Does the Premise *entail* or *contradict* the Hypothesis?

E: C:

- Disagreement in annotation is not noise, but information (Aroyo and Welty 2015)
- HLV used a tool to improve the model's performance/fairness

Lora Aroyo and Chris Welty. 2015. Truth is a lie: Crowd truth and the seven myths of human annotation. AI Magazine, 36(1):15–24.

RLHF - preference training

Content Preference HLV

Prompt : Would embodiment of an LLM be a prerequisite for sentience?

Response A: No, embodiment of an LLM would not be a prerequisite for sentience ...

Response B: The relationship between embodiment and sentience in LLMs remains an open question ...

Annotator 1: Prefers B **Annotator 2** : Prefers A

Figure 2. **HLV**examples from the preference dataset MutliPref

NLP tasks: From tame to **wicked** problem

Stage	Human Adjacent	Human Aware	Human Centered
Tasks	Expert systems, NER, POS, etc (Ashley, 1991)	Everyday language use: Sentimental Analysis, hate speech detection, etc (Pérez-Rosas et al., 2018)	Interactive and creative open-ended tasks: pair programming, creative writing, counseling (Litschko et al., 2023)

https://aclanthology.org/2026.findings-acl.1190/

NLP tasks: From tame to **wicked** problem

Tame problems

- Have a relatively clear problem definition and objective.
- Success can be measured with agreed-upon criteria.
- Solutions can be tested and compared.

Wicked problems

- Have no single agreed-upon definition of the problem or the “correct” solution.
- Involve multiple stakeholders with different values and goals.
- Solving one aspect may create new problems elsewhere.
- The problem itself changes as technology and society change.

Tokenisation & Sentence Splitting
Text Classification (slides)

6. Oct. 2026:
Information Extraction (slides)
Question Answering (slides)

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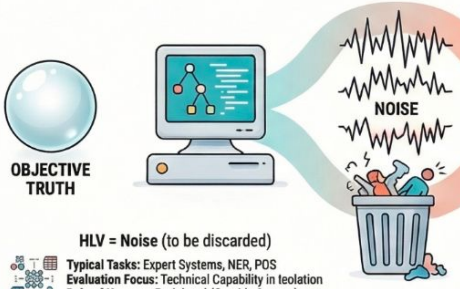
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Evaluation	**Capability**: technical components in isolation (Jones and Galliers, 1995)	**Human Interaction**: bias, fairness and interpretability (Belinkov et al., 2020; Mehrabi et al., 2021)	**Systemic impact**: broader impacts on the social systems (Weidinger et al., 2023)
Role of Humans	**Peripheral**: largely outside the system's	**Acknowledged**: humans as data annotator and end-users	**Central**: humans as co-creators
View of HLV	***Noise*** to be discarded	***Signal*** to improve performance	***Selbstzweck*** to align diverse human values

A Roadmap for Human-Centred AI Alignment, Valuing Human Pluralism as an Intrinsic End

STAGE 1: HUMAN ADJACENT
(Pre-1990s)

HLV = Noise (to be discarded)

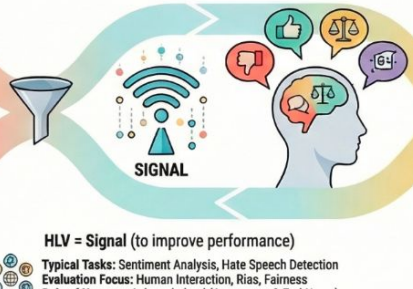


HLV = Noise (to be discarded)

Typical Tasks: Expert Systems, NER, POS
Evaluation Focus: Technical Capability in teolation
Role of Humans: Peripheral (Outside System)

STAGE 2: HUMAN AWARE
(1990s – 2022)

HLV = Signal (to improve performance)

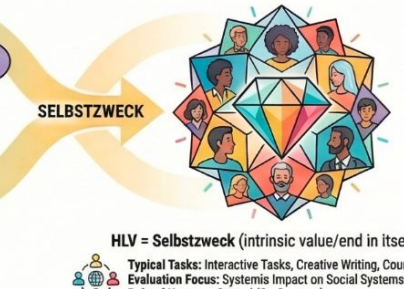


HLV = Signal (to improve performance)

Typical Tasks: Sentiment Analysis, Hate Speech Detection
Evaluation Focus: Human Interaction, Rias, Fairness
Role of Humans: Acknowledged (Annotators & End-Users)

STAGE 3: HUMAN CENTRED
(2023 – Present)

HLV = Selbstzweck (intrinsic value/end in itself)



HLV = Selbstzweck (intrinsic value/end in itself)

Typical Tasks: Interactive Tasks, Creative Writing, Counselling
Evaluation Focus: Systemis Impact on Social Systems
Role of Humans: Centrol (Co-Creators)

ACTIONABLE STRATEGIES FOR PLURALISTIC DATASETS

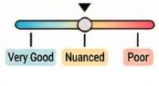
Diversify the Annotator Pool



Release Annotator-Level Preferences



Leverage Heterogeneous Feedback

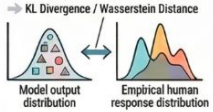


EXAMPLE:
Pluralism as Reasoning

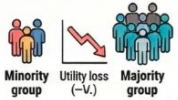


FUTURE EVALUATION METRICS FOR ALIGNMENT

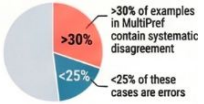
DEFINITION:
Pluralism Fidelity



DEFINITION:
Minority Regret



SUPPORTING FACT: **Systematic Disagreement is Common**



Human Label Variation (HLV)

- In the LLM era
- more personal, interactive and creative tasks, more social-political controversial contexts
- Psychology Counselling, Political Voting Assistant
- Important for the fairness/reliability/ of AI
- HLV captures Pluralistic human values (PHV) /opinions
- PLV as a value itself (Selbstzweck) (~~noise, information~~)



Through the Lens of Split Vote: Exploring Disagreement, Difficulty and Calibration in Legal Case Outcome Classification

Shanshan Xu[1,2], **Santosh T.Y.S.S**[1], **Oana Ichim**[3]
Barbara Plank[4,5], **Matthias Grabmair**[1]
[1]Technical University of Munich, Germany [2]ELTEMATE
[3]Graduate Institute of International and Development Studies, Switzerland
[4]IT University of Copenhagen, Denmark
[5]LMU Munich & Munich Center for Machine Learning (MCML), Germany

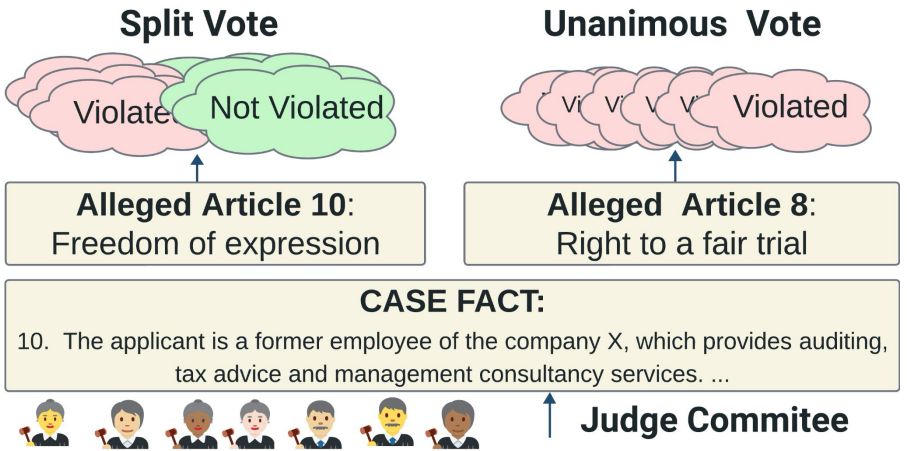






Case Outcome Classification & Split Vote

- **Case Outcome Classification** (COC): classification of legal case outcomes based on textual descriptions of their facts.
- **Split Votes** (SV): judges cannot reach a unanimous decision
- SV cases are often controversial and politically salient



Motivation

- Currently COC dataset has only one single label, which is the majority vote of multiple judges
- SV is a naturally occurred and observable human label variation (HLV), which mirrors the diverse and pluralistic nature of human values

Reliability

Overall reliability of a model is determined by two sides:

1) Trustworthiness: **performance alignment** - The model should know when it doesn't know

2) Fairness: **human alignment** - The model shouldn't be over certain, when human are not certain.

Our Contribution

- **SV-ECHR**: A case outcome classification (COC) dataset with SV information in European Court of Human Rights(ECtHR)
- **A taxonomy of disagreement**: analyze the reasons behind judges' split votes
- Investigation on the **alignment between models and humans**. We observe limited human alignment with the judge vote distribution.

Disagreement Taxonomy

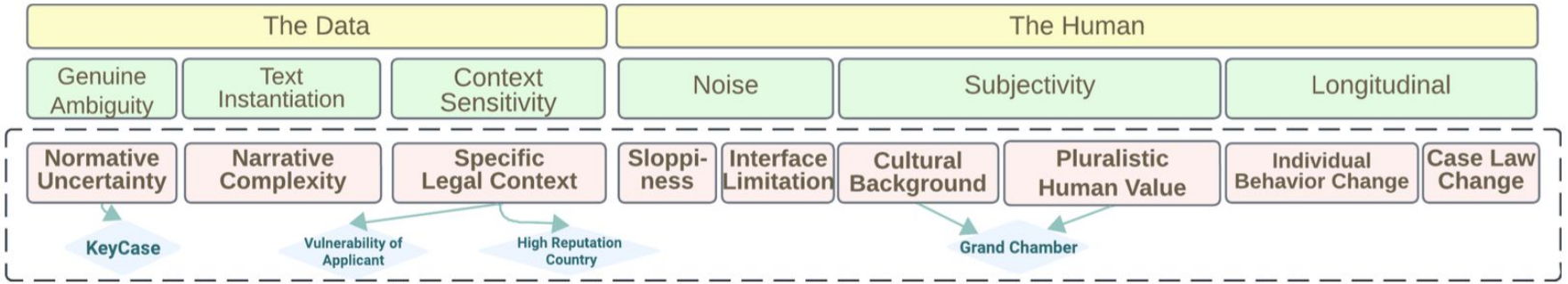


Figure 2: Taxonomy of disagreement sources among judges, with two adapted task-agnostic levels from Xu et al. 2023b and our expanded split-vote specific subcategories and proxy variables (in dashed box).

Dataset: SV-ECHR

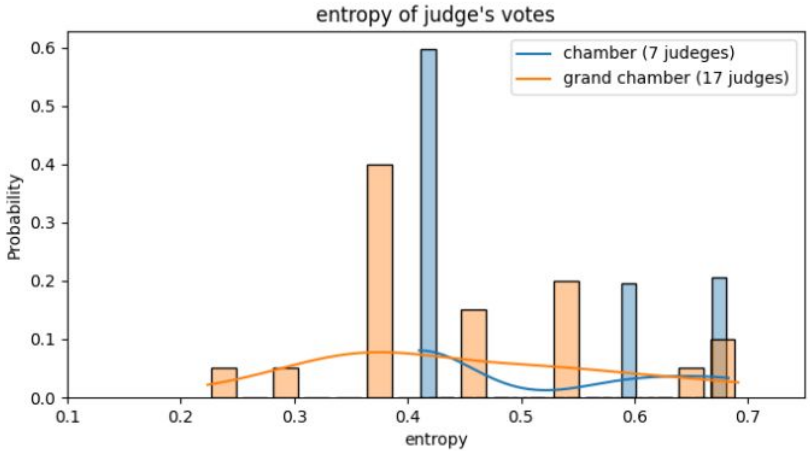


Figure 6: Histogram of entropy of judges' votes distribution over train/dev/test split

A large share (~60%) of Judges' votes are *single dissenting votes*, where only one single judge voted differently than the other judge majority.

COC Performance

Task: Multilabel Classification. (ECtHR A|B; Violation Identification given Allegation) identifies the set of violated convention article(s) from the textual case facts and list of allegedly violated articles.

Models: hierarchical models of LegalBERT/BERT

Metrics: mic-F1, mac-F1, and hard-macro-F1 (hm-F1). The hm-F1 is calculated as the mean F1-score for each article, where cases with that article having been violated are considered as positive instances, and cases with that article being alleged but not found to be violated as negative instances, resulting in a smaller pool of more difficult negatives.

	mic-F1	mac-F1	hm-F1
BERT	72.31 ± 4.09	65.53 ± 6.22	64.22 ± 3.46
LegalBERT	73.55 ± 0.62	68.75 ± 1.13	68.03 ± 1.29

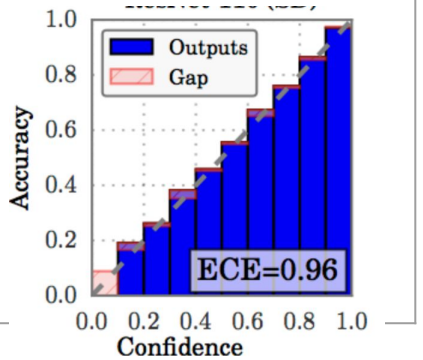
Confidence calibration

- The model should be unsure when it does not know the answer.
- A model is considered well confidence-calibrated if its prediction confidence aligns with its predictive accuracy, commonly evaluated against the human majority class.

Human calibration

- The model should be unsure when humans are unsure about the answer.
- A model well human-calibrated if the categorical distribution of predicted class probabilities align well with the actual human vote distribution.

Confidence Calibration

Calibration Methods	**Temperature Scaling**(TS; Guo et al 2017) post-hoc method uses a single temperature parameter t to scale the output logits of a classifier	`softmax = e^(z/T) / sum_i e^(z_i/T)`
Metrics	**Expected Calibration Error** (ECE; Naeini et al. 2015): A lower ECE indicates better calibration, suggesting that the model's predicted probabilities are more accurate reflections of the true probabilities.	$\mathrm{ECE} = \sum_{m=1}^{M} \frac{\lvert B_m \rvert}{N} \lvert \mathrm{acc}\,(B_m) - \mathrm{conf}\,(B_m) \rvert$ 

Guo, Chuan, et al. "On calibration of modern neural networks." *International conference on machine learning*. PMLR, 2017.
Naeini, Mahdi Pakdaman, Gregory Cooper, and Milos Hauskrecht. "Obtaining well calibrated probabilities using bayesian binning." *Proceedings of the AAAI conference on artificial intelligence*. Vol. 29. No. 1. 2015.

MODEL CALIBRATION IN NLP

Expected Calibration Error

When a model says 80%, should we believe it?



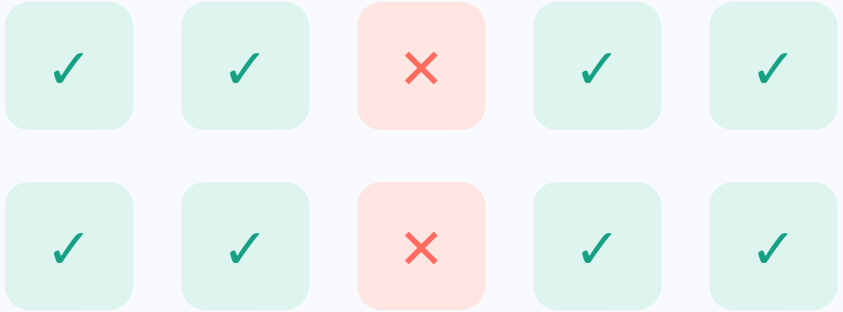
For CS master’s students

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CORE INTUITION

Calibration connects confidence to observed accuracy

Collect predictions made near 80% confidence



8 correct out of 10

A calibrated model

Among all predictions with roughly 80% confidence, about 80% should be correct.

Confidence answers: How sure is the model?
Accuracy answers: How often was it right?

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DEFINITION

Expected Calibration Error

Split predictions into M confidence bins



For each bin B_m

$\text{acc}(B_m)$ = fraction correct
$\text{conf}(B_m)$ = mean confidence

$$\text{ECE} = \Sigma_m \ |B_m| / n \cdot |\text{acc}(B_m) - \text{conf}(B_m)|$$

Weighted average of the confidence–accuracy gaps

generated with CodeX

CALCULATION

Worked example: ECE = 8.5%

Confidence bin	Predictions	Mean confidence	Accuracy	Absolute gap
50–70%	20	60%	50%	10%
70–90%	50	80%	70%	10%
90–100%	30	95%	90%	5%

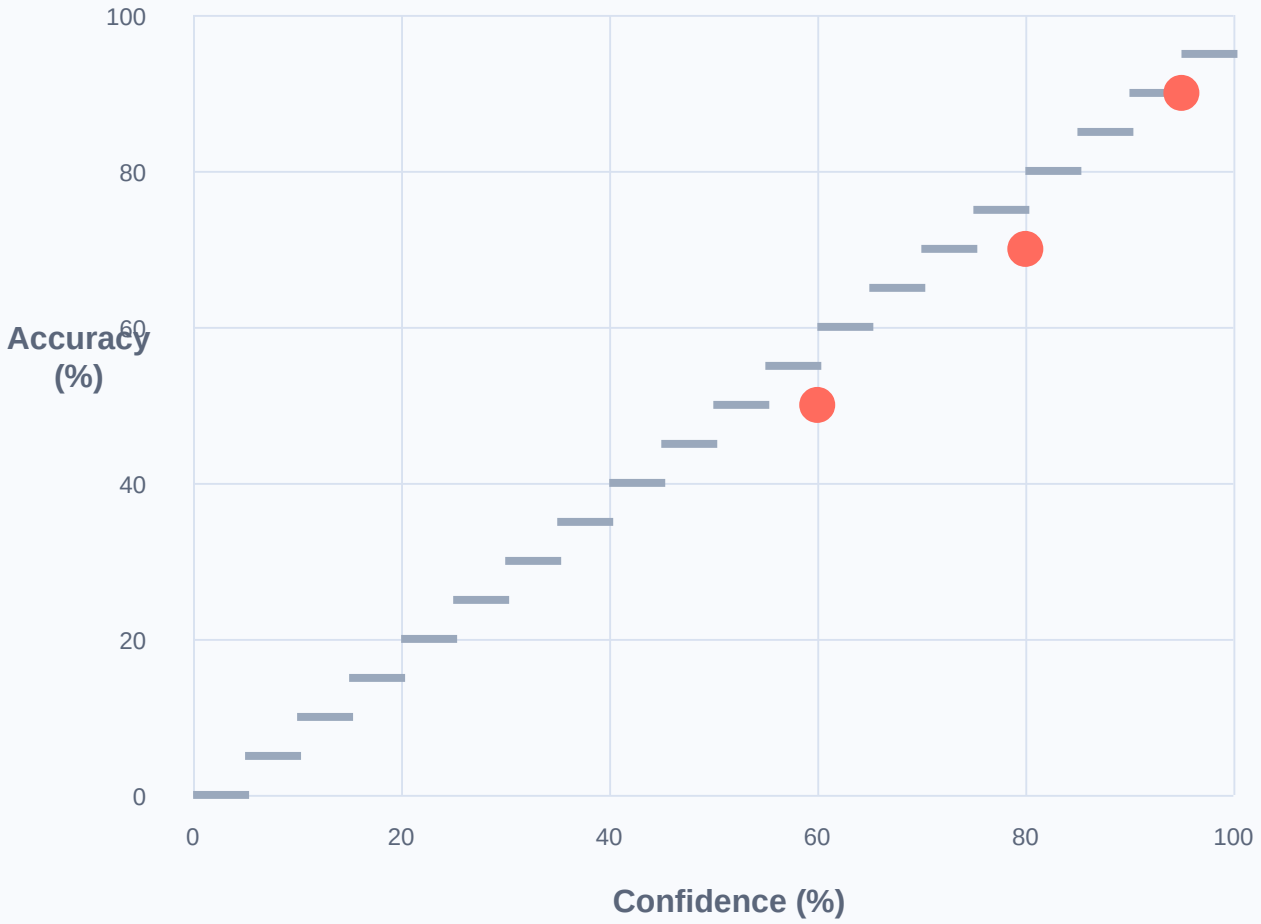
ECE = 0.20×0.10 + 0.50×0.10 + 0.30×0.05 = 0.085

The model’s average calibration gap is 8.5 percentage points under this binning scheme.

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DIAGNOSIS

Reliability diagram



Perfect calibration

Points lie on the diagonal.

Below the diagonal

Confidence exceeds accuracy.
The model is overconfident.

Above the diagonal means underconfidence.

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APPLICATIONS

Why calibration matters in NLP systems

Human review

Route uncertain legal or medical predictions to specialists.

Selective prediction

Answer only when estimated reliability clears a threshold.

Information extraction

Use confidence to filter entities, relations, or labels.

System composition

Pass meaningful probabilities into later decision stages.

Accuracy measures correctness. Calibration measures whether confidence is trustworthy.

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INTERPRETATION

ECE needs context

What ECE can hide

- Bin choices change the value
- Sparse bins create noisy estimates
- One number can hide class or subgroup failures
- Standard ECE uses only the top-class confidence

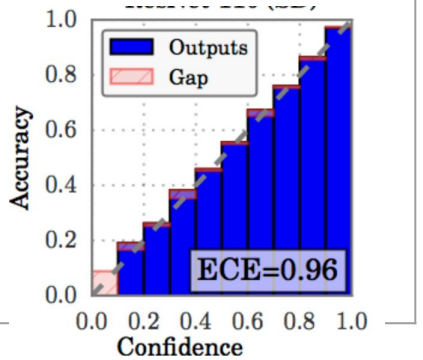
Good reporting practice

- ✓ Report accuracy or F1 beside ECE
- ✓ Show a reliability diagram
- ✓ Inspect classes and important subgroups
- ✓ Test again after domain shift

Low ECE does not imply high accuracy. A weak model can still be calibrated.

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Human Calibration

Calibration Methods	**Soft Loss Function** (Peterson et al.) “soft labels” derived from the judges’ vote distributions	$-\sum_{i=1}^{n}\sum_{c} p_{hum}\left(y_i \mid x_i\right) \log p_\theta\left(y_i = c \mid x_i\right.$
Metrics	**DistCE**(Baan et al. 2022): measures the difference of vote distribution q and model’s predictive distribution p	DistCE(x) = 1/2 ||q - p||

Joshua C. Peterson, Ruairidh M. Battleday, Thomas L. Griffiths, and Olga Russakovsky. 2019. Human un- certainty makes classification more robust. (ICCV)
Joris Baan, Wilker Aziz, Barbara Plank, and Raquel Fernandez. 2022. Stop measuring calibration when humans disagree. EMNLP

Results & Discussion

		hm-F1 ↑			ECE ↓			DistCE ↓			coun
		/	TS	soft	/	TS	soft	/	TS	soft	
LegalBERT	u	**69.30 ± 1.88**	**69.30 ± 1.88**	67.31 ± 0.5	23.32 ± 1.01	**2.95 ± 0.67**	22.04 ± 0.63	25.10 ± 1.17	37.23 ± 1.71	**24.70 ± 0.17**	1463
	sv	**53.67 ± 4.59**	**53.67 ± 4.59**	46.21 ± 4.41	29.92 ± 2.81	**8.02 ± 0.58**	28.49 ± 0.82	41.03 ± 2.75	**28.28 ± 0.99**	40.72 ± 1.00	112
	all	**68.03 ± 1.29**	**68.03 ± 1.29**	65.16 ± 0.82	23.75 ± 1.08	**2.99 ± 0.83**	22.32 ± 0.64	26.23 ± 1.0	36.60 ± 1.52	**25.84 ± 0.21**	1575

Table 3: COC performance (hm-F1), confidence-calibration (ECE), and human-calibration (DistCE) performan with std (±) on test set. "/": COC finetuned models in § 6.1; "soft": models fine-tuned with soft-loss; "TS": mod

- Without Calibration: SV cases are indeed most challenging: lower COC performance (hm-F1), higher degree of miscalibration (higher ECE) and misalignment with human responses (higher distCE).

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- Confidence calibration: applying TS greatly reduces ECE, without negatively impacting hm-F1, enhancing confidence calibration

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- Human calibration: models remain misaligned with human vote distribution as the DistCE scores remain high
- Soft-loss tuning slightly improves the alignment between the model's prediction confidence and the human vote distribution (lower DistCE)
- TS models exhibit substantially lower DistCE scores on split-vote cases than soft models.

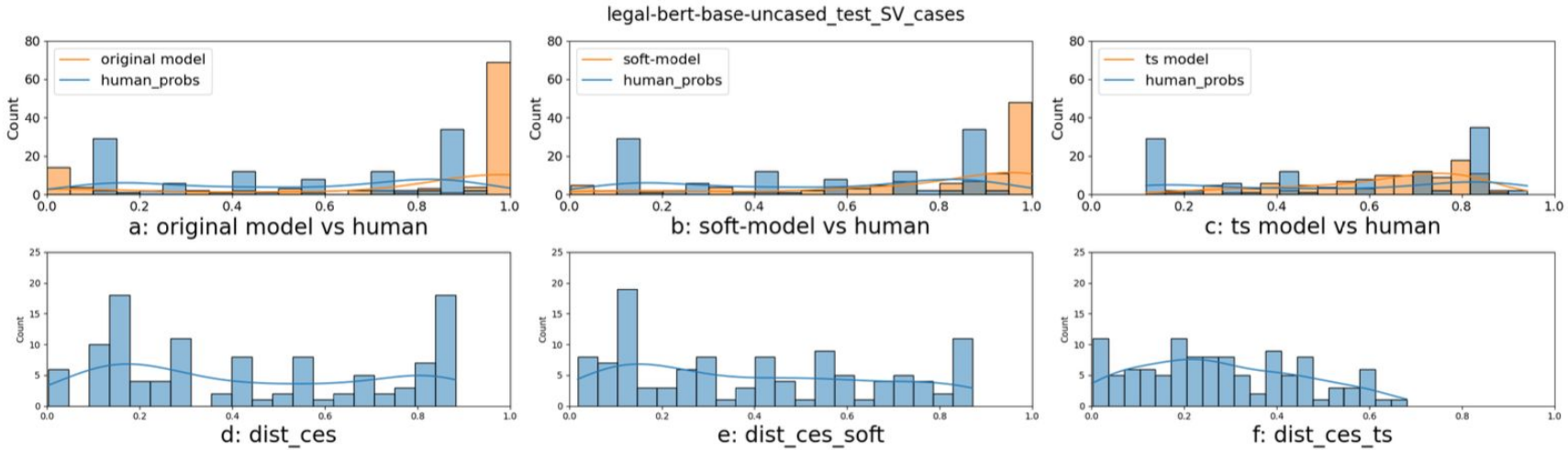


Figure 3: The distributions over probabilities for class 1 of the models vs human vote distributions (row 1) and distCE (row 2). See Fig 8 in App S for more figures comparing human uncertainty to model uncertainty

- The overly aggressive temperature, resulting in a more uniform output distribution (fig c)
- judge votes often exhibit a quasi-bimodal distribution, with many split votes caused by a single judge dissenting pro/con the finding of a violation
- TS has fewer perfectly human-alligned instances (DistCE scores below 0.15)

Conclusion

- Dataset SV-ECHR and disagreement taxonomy
- Our Experiments reveals:
  - Soft loss training produces only slightly better human calibration scores
  - Shortcomings when combining TS and ECE to improve and measure calibration of models against labels subject to inherent human disagreement. This is due to a distribution mismatch reflected in low DistCE scores.
- We call on the community to explore methods for measuring and improving the alignment of model calibration with human behavior